Q1. JEE Main 2026 (21 January Shift 2)

Let $y = y(x)$ be the solution of the differential equation $\sec x \frac{dy}{dx} - 2y = 2 + 3\sin x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), y(0) = -\frac{7}{4}$. Then $y\left(\frac{\pi}{6}\right)$ is equal to :

(1) $-\frac{5}{4}$
(2) $-\frac{5}{2}$
(3) $-3\sqrt{3} - 7$
(4) $-3\sqrt{2} - 7$

Q2. JEE Main 2026 (23 January Shift 1)

Let $y = y(x)$ be the solution of the differential equation $x^4\, dy + \left(4x^3 y + 2\sin x\right)dx = 0, x > 0, y\left(\frac{\pi}{2}\right) = 0$. Then $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to :

(1) 72
(2) 92
(3) 64
(4) 81

Q3. JEE Main 2026 (23 January Shift 1)

Let f be a twice differentiable non-negative function such that $(f(x))^2 = 25 + \int_0^x \left((f(t))^2 + (f'(t))^2\right)dt$. Then the mean of $f(\log_e(1)), f(\log_e(2)), \ldots\ldots, f(\log_e(625))$ is equal to ____.

Q4. JEE Main 2026 (28 January Shift 2)

Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} - y = x^2 \cot x, x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then $6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$ is equal to :

(1) 3π
(2) π
(3) $-\pi$
(4) -3π

Q5. JEE Main 2026 (24 January Shift 2)

Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$, and $\lim_{t \to x}\left(\frac{t^2 y(x) - x^2 y(t)}{x - t}\right) = 3$ for each $x > 0$. Then $2y(2)$ is equal to

(1) 23
(2) 27
(3) 12
(4) 18

Q6. JEE Main 2026 (22 January Shift 2)

If $y = y(x)$ satisfies the differential equation $16(\sqrt{x + 9\sqrt{x}})(4 + \sqrt{9 + \sqrt{x}})\cos y\, dy = (1 + 2\sin y)dx, x > 0$ and $y(256) = \frac{\pi}{2}, y(49) = \alpha$, then $2\sin\alpha$ is equal to :

(1) $2(\sqrt{2} - 1)$
(2) $3(\sqrt{2} - 1)$
(3) $2\sqrt{2} - 1$
(4) $\sqrt{2} - 1$

Q7. JEE Main 2026 (22 January Shift 1)

Let the solution curve of the differential equation $xdy - ydx = \sqrt{x^2 + y^2}dx, x > 0, y(1) = 0$, be $y = y(x)$. Then $y(3)$ is equal to

(1) 1
(2) 4
(3) 2
(4) 6

Q8. JEE Main 2026 (22 January Shift 1)

Let $f:[1,\infty)\to\mathbb{R}$ be a differentiable function. If $6\int_1^x f(t)dt = 3xf(x)+x^3-4$ for all $x\geq 1$, then the value of $f(2)-f(3)$ is

(1) -4 (2) 3 (3) 4 (4) -3

Q9. JEE Main 2026 (23 January Shift 2)

If the solution curve $y=f(x)$ of the differential equation
$(x^2-4)y'-2xy+2x(4-x^2)^2=0, x>2,$
passes through the point $(3,15)$, then the local maximum value of f is ____.

Q10. JEE Main 2026 (21 January Shift 1)

Let $y=y(x)$ be the solution curve of the differential equation $(1+x^2)\mathrm{d}y+(y-\tan^{-1}x)dx=0, y(0)=1$. Then the value of $y(1)$ is :

(1) $\frac{4}{e^{\pi/4}}+\frac{\pi}{2}-1$ (2) $\frac{2}{e^{\pi/4}}+\frac{\pi}{4}-1$
(3) $\frac{2}{e^{\pi/4}}-\frac{\pi}{4}-1$ (4) $\frac{4}{e^{\pi/4}}-\frac{\pi}{2}-1$

Q11. JEE Main 2026 (28 January Shift 1)

Let $y=y(x)$ be the solution of the differential equation $x\frac{dy}{dx}-\sin 2y = x^3(2-x^3)\cos^2 y, x\neq 0$. If $y(2)=0$, then $\tan(y(1))$ is equal to

(1) $-\frac{3}{4}$ (2) $\frac{3}{4}$ (3) $-\frac{7}{4}$ (4) $\frac{7}{4}$

ANSWERS AND SOLUTIONS

1. (2)	**2.** (4)	**3.** 1565	**4.** (3)	**5.** (1)	**6.** (3)	**7.** (2)	**8.** (2)
9. 16	**10.** (2)	**11.** (4)					

1. **(2)** The differential equation is $\sec x \frac{dy}{dx} - 2y = 2 + 3\sin x$, or $\frac{dy}{dx} - 2\cos x \cdot y = (2 + 3\sin x)\cos x$.
This is a linear ODE with integrating factor $\mu(x) = e^{-2\sin x}$.
Multiplying both sides and integrating: $\frac{d}{dx}[ye^{-2\sin x}] = (2 + 3\sin x)\cos x \cdot e^{-2\sin x}$
Computing the integral: $\int 2\cos x \cdot e^{-2\sin x} dx = -e^{-2\sin x}$
For the second term, using integration by parts: $\int 3\sin x \cos x \cdot e^{-2\sin x} dx = \frac{3}{4}e^{-2\sin x}(-2\sin x - 1)$
Thus: $ye^{-2\sin x} = -e^{-2\sin x} + \frac{3}{4}e^{-2\sin x}(-2\sin x - 1) + C$
Simplifying: $y = -\frac{7}{4} - \frac{3}{2}\sin x + Ce^{2\sin x}$
Using $y(0) = -\frac{7}{4}$ gives $C = 0$.
Therefore: $y(\frac{\pi}{6}) = -\frac{7}{4} - \frac{3}{2} \cdot \frac{1}{2} = -\frac{5}{2}$

2. **(4)** $x^4\, dy + 4x^3 y\, dx + 2\sin x\, dx = 0 \Rightarrow d(x^4 y) + 2\sin x\, dx = 0$.
Integrating: $x^4 y - 2\cos x = C$.
$y(\pi/2) = 0$: $(\pi/2)^4 \cdot 0 - 2\cos(\pi/2) = 0 \Rightarrow C = 0$.
$x^4 y = 2\cos x$.
At $x = \pi/3$: $(\pi/3)^4 y(\pi/3) = 2\cos(\pi/3) = 1$.
$\pi^4 y(\pi/3) = 81 \cdot (\pi/3)^4 y(\pi/3) \cdot \frac{1}{(\pi/3)^4} \cdot \frac{\pi^4}{81}$... Directly: $y(\pi/3) = \frac{81}{\pi^4}$, so $\pi^4 y(\pi/3) = 81$.

3. **(1565)** Differentiating $(f(x))^2 = 25 + \int_0^x ((f(t))^2 + (f'(t))^2)\, dt$:
$2ff' = f^2 + (f')^2 \Rightarrow (f - f')^2 = 0 \Rightarrow f' = f$.
$f(x) = Ce^x$. At $x = 0$: $f(0)^2 = 25 \Rightarrow f(0) = 5$ (non-negative), so $f(x) = 5e^x$.
$f(\log_e k) = 5k$ for $k = 1, 2, \ldots, 625$.
Mean $= \frac{5(1+2+\cdots+625)}{625} = \frac{5\times625\times626}{2\times625} = \frac{5\times626}{2} = 1565$.

4. **(3)** The differential equation $x\frac{dy}{dx} - y = x^2 \cot x$ can be rewritten as $\frac{dy}{dx} - \frac{y}{x} = x\cot x$.
The integrating factor is $\mu(x) = \frac{1}{x}$. Multiplying by $\mu(x)$ gives $\frac{d}{dx}\left(\frac{y}{x}\right) = \cot x$.
Integrating: $\frac{y}{x} = \ln|\sin x| + C$, so $y = x\ln(\sin x) + Cx$.
Using $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$: we get $\frac{\pi}{2} = 0 + C\frac{\pi}{2}$, thus $C = 1$.
Therefore $y = x(1 + \ln(\sin x))$.
Now: $y\left(\frac{\pi}{6}\right) = \frac{\pi}{6}(1 - \ln 2)$ and $y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}\left(1 - \frac{\ln 2}{2}\right)$.
Thus $6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right) = \pi(1 - \ln 2) - 2\pi\left(1 - \frac{\ln 2}{2}\right) = -\pi$.

5. **(1)** Given the limit $\lim_{t\to x} \frac{t^2 y(x) - x^2 y(t)}{x - t} = 3$.
The limit is in $\frac{0}{0}$ form. Applying L'Hopital's Rule with respect to t:

$\lim_{t\to x} \frac{\frac{d}{dt}(t^2y(x)-x^2y(t))}{\frac{d}{dt}(x-t)} = 3$

$\lim_{t\to x} \frac{2ty(x)-x^2y'(t)}{-1} = 3$

Substituting $t = x$:

$\frac{2xy(x)-x^2y'(x)}{-1} = 3$

$x^2y'(x) - 2xy(x) = 3$

Dividing by x^4 to make it a linear differential equation or recognizing the quotient rule form:

$\frac{x^2y'(x)-2xy(x)}{x^4} = \frac{3}{x^4}$

$\frac{d}{dx}\left(\frac{y(x)}{x^2}\right) = \frac{3}{x^4}$

Integrating both sides with respect to x:

$\frac{y(x)}{x^2} = \int 3x^{-4}dx = \frac{3x^{-3}}{-3} + C = -\frac{1}{x^3} + C$

Given $y(1) = 2$:

$\frac{2}{1^2} = -\frac{1}{1^3} + C \implies 2 = -1 + C \implies C = 3$

So, $\frac{y(x)}{x^2} = -\frac{1}{x^3} + 3$

$y(x) = -\frac{1}{x} + 3x^2$

We need to find $2y(2)$:

$y(2) = -\frac{1}{2} + 3(2^2) = -\frac{1}{2} + 12 = \frac{23}{2}$

$2y(2) = 2 \times \frac{23}{2} = 23$

6. **(3)** $\int \frac{\cos y}{1+2\sin y}dy = \int \frac{dx}{16(\sqrt{9\sqrt{x}+x})(4+\sqrt{9+\sqrt{x}})}$

$4 + \sqrt{9+\sqrt{x}} = t$

$\frac{1}{2\sqrt{9+\sqrt{x}}} \times \frac{dx}{2\sqrt{x}} = 1dx$

$\frac{1}{2}\ell n|1 + 2\sin y| = \int \frac{4dt}{16t} + C$

$\frac{1}{2}\ell n|1 + 2\sin y| = \frac{1}{4}\ell n\left|\ 4 + \sqrt{9+\sqrt{x}} + C\right.$

$\frac{1}{2}\ell n(2\sin y + 1) = \frac{1}{4}\ell n|4 + \sqrt{9+\sqrt{x}}| + C$

Substituting $\left(256, \frac{\pi}{2}\right)$

$\frac{1}{2}\ell n3 = \frac{1}{2}\ell n3 + C \quad C = 0$

Substituting $(49, \alpha)$

$\frac{1}{2}\ell\text{n}(2\sin\alpha + 1) = \frac{1}{4}\ell\text{n}8$

$\ell\text{n}(2\sin\alpha + 1) = \frac{1}{2}\ell\text{n}8$

$\ell\text{n}(2\sin\alpha + 1) = \ell\text{n}2\sqrt{2}$

$2\sin\alpha + 1 = 2\sqrt{2}$

$2\sin\alpha = 2\sqrt{2} - 1$

7. **(2)** $xdy - ydx = \sqrt{x^2+y^2}\,dx$.
Put $y = vx$: $x^2dv = x\sqrt{1+v^2}\,dx$.
$\frac{dv}{\sqrt{1+v^2}} = \frac{dx}{x}$.
Integrating: $\ln(v+\sqrt{1+v^2}) = \ln x + C$.
$y(1) = 0 \Rightarrow v = 0$ at $x = 1$: $C = 0$.
$v + \sqrt{1+v^2} = x \Rightarrow y + \sqrt{x^2+y^2} = x^2$.
At $x = 3$: $y + \sqrt{9+y^2} = 9$
$\Rightarrow 9 + y^2 = (9-y)^2$
$\Rightarrow 18y = 72$
$\Rightarrow y = 4$.

8. **(2)** Differentiating $6\int_1^x f(t)dt = 3xf(x) + x^3 - 4$:
$6f(x) = 3f(x) + 3xf'(x) + 3x^2 \Rightarrow f(x) - xf'(x) = x^2$.
$\frac{d}{dx}\left(\frac{f(x)}{x}\right) = \frac{xf'(x)-f(x)}{x^2} = -1$.
Integrating: $\frac{f(x)}{x} = -x + C \Rightarrow f(x) = -x^2 + Cx$.
At $x = 1$: $0 = 3f(1) - 3 \Rightarrow f(1) = 1$. So $-1 + C = 1 \Rightarrow C = 2$.
$f(x) = -x^2 + 2x$. $f(2) = 0$, $f(3) = -3$.
$f(2) - f(3) = 3$.

9. **(16)** Rewriting: $\frac{dy}{dx} - \frac{2x}{x^2-4}y = -2x(x^2-4)$.
IF $= e^{-\int \frac{2x}{x^2-4}dx} = \frac{1}{x^2-4}$.
$\frac{d}{dx}\left(\frac{y}{x^2-4}\right) = -2x \Rightarrow \frac{y}{x^2-4} = -x^2 + C$.
$y = (x^2-4)(C-x^2)$. At $(3, 15)$: $15 = 5(C-9) \Rightarrow C = 12$.
$y = (x^2-4)(12-x^2)$.
$y' = 4x(8-x^2) = 0 \Rightarrow x = 2\sqrt{2}$ (for $x > 2$).
$y'' = 32 - 12x^2 = -64 < 0$ at $x = 2\sqrt{2}$, confirming local maximum.
$f(2\sqrt{2}) = (8-4)(12-8) = 16$.

10. **(2)** Rewrite as $\frac{dy}{dx} + \frac{y}{1+x^2} = \frac{\tan^{-1}x}{1+x^2}$.
IF $= e^{\tan^{-1}x}$.
$y \cdot e^{\tan^{-1}x} = \int \frac{\tan^{-1}x}{1+x^2} \cdot e^{\tan^{-1}x}dx$.
Let $t = \tan^{-1}x$: $= \int te^t dt = e^t(t-1) + C$.
$y = \tan^{-1}x - 1 + Ce^{-\tan^{-1}x}$.
Using $y(0) = 1$: $C = 2$.
$y(1) = \frac{\pi}{4} - 1 + \frac{2}{e^{\pi/4}}$.

11. **(4)** Divide the equation by $\cos^2 y$:
$x\sec^2 y\frac{dy}{dx} - 2\tan y = x^3(2-x^3)$.

Let $v = \tan y$, so $\sec^2 y \frac{dy}{dx} = \frac{dv}{dx}$.
The equation becomes $x\frac{dv}{dx} - 2v = x^3(2 - x^3)$.
Dividing by x: $\frac{dv}{dx} - \frac{2v}{x} = x^2(2 - x^3)$.
Using integrating factor $\mu = x^{-2}$: $\frac{d}{dx}(x^{-2}v) = 2 - x^3$.
Integrating: $x^{-2}v = 2x - \frac{x^4}{4} + C$.
From $y(2) = 0$, we have $\tan 0 = 0 = 16 - 16 + 4C$, giving $C = 0$.
Therefore $\tan y = 2x^3 - \frac{x^6}{4}$.
At $x = 1$: $\tan(y(1)) = 2 - \frac{1}{4} = \frac{7}{4}$.